COMP 4107 FINAL PROJECT PROPOSAL (Group 5 – Ajaan Nalliah, Vipra Gajera, James Yap)

Background & Objectives

Prostate Cancer is one of the most commonly diagnosed cancers among men, and Intraductal Carcinoma of the Prostate (IDC-P) is an aggressive subtype of Prostate Cancer which could be associated with poor clinical outcomes. Accuracy in diagnosis is crucial; however, it can vary among pathologists. Hence, resulting in the motivated use of automated classification methods through imaging data.

The recent work by the LAMPE Lab at Carleton University (Gagnon et al., 2025) utilized multimodal nonlinear optical (NLO) imaging paired with the Grey-Level Co-occurrence Matrix (GLCM) texture features and Support Vector Machine (SVM) in order to classify prostate tissue into four categories: IDC-P, high-grade cancer, low-grade cancer, and benign tissue. Their approach achieved up to 98% accuracy at the field-of-view (FOV) level. However, first-order pixel statistics were found to be nearly stochastic, indicating that spatial texture patterns are crucial discriminative features.

The objective of this project is to investigate whether Convolutional Neural Networks (CNNs), through transfer learning, can learn spatial texture representations compared with handcrafted GLCM features. We aim to:

1. Compare CNN classification performance with the provided GLCM + SVM baseline from the paper.
2. Evaluate the multimodal fusion of imaging channels to improve the classification accuracy.
3. Using the Grad-CAM visualization to comprehend the spatial features CNN uses for classification.

This determines deep learning's ability to accurately and interpretably classify prostate cancer tissue.

Proposed Methods from Artificial Intelligence

Using a confusion matrix for validation across classes. Implementing the transfer learning through pretrained ImageNet architectures such as ResNet-50 and EfficientNet-B3. These architectures will be modified by replacing the final fully connected layer with a new classification layer with four output neurons (i.e. the four tissue classes).

Each microscopy image sample consists of three aligned imaging modalities: SHG (Collagen), 1668 cm^(-1) SRS (protein), 1450 cm^(-1) SRS (lipids). These will combine into a three-channel input tensor with the size: input is an element of R^(224 x 224 x 3), to allow the CNN to learn multimodal spatial representations. Experimenting with the number of kernels on each layer to identify the best number of intermediate channel depths between layers, in addition to kernel sizes and strides.

The training will take place through supervised learning through CCE (Categorical-Cross Entropy) Loss: L = -$\Sigma y_i log(\widehat{y_i})$, where y_i is the true label and $\widehat{y}_i$ is the predicted class probability. The next step will be to perform optimization through stochastic gradient descent with momentum or Adam optimizer. Transfer learning by initially freezing pretrained convolutional layers and training the classifier head, and then fine-tuning the deep convolutional layers. We can conduct data augmentation to improve the generalization, including: rotation, horizontal and vertical flipping, translation, and random cropping. It will help the network learn better rotationally and translationally invariant texture representations. The evaluation metrics (will be) used for the model: Classification accuracy, confusion matrix, precision, recall, and F1-score, patient-level cross-validation accuracy.

Dataset or Environment Used

The authors of the original paper (Gagnon et al., 2025) and researchers at the Laser-Assisted Medical Physics and Engineering (LAMPE) lab include Prof. Sangeeta Murugkar and PhD students Justin Gagnon and Parminder Riarh from the Carleton University Department of Physics. We have collaborated with the research team to obtain guidance and access to the microscopy images used.

The dataset includes labelled images of IDC-P, LGC, HGC, and benign prostate tissue samples from a cohort of 50 patients who participated in the Canadian Prostate Genome Project (CPC-GENE). This project utilizes co-registered Stimulated Raman Scattering (SRS) and Second Harmonic Generation (SHG) microscopy to provide complementary biochemical and structural data. While SRS images lipids at 1450 cm^{-1} and proteins at 1668 cm^{-1}, SHG specifically maps non-centrosymmetric structures like collagen fibers. For all three of these channels, first order statistics such as mean, variance, skewness and kurtosis, as well as second order statistics, i.e., contrast, correlation, energy, homogeneity and entropy were computed. Manual feature selection was then performed by means of the Kruskal-Wallis test to show the discriminatory power of these statistical values over different cell types. Finally, majority-consensus voting was used as an ensemble approach for consolidating multimodal inference. (For our approach, we aim to construct a model that will self-learn these features with minimal manual intervention.)

Additionally, the dataset includes data point identifiers such as microscopy slide numbers, core positions, cell classes, and Field of View (FOV) identifiers. This information is crucial for eliminating intra-core bias during the validation of inference methodologies, as it allows for the discrimination of samples at the patient level. To increase the number of data points, larger FOV images of $70\ \mu m \times 70 \mu m$ were split into 3×3 grids, resulting in 267 IDC-P subimages, 544 HGC subimages, 221 LGC subimages, and 46 benign tissue subimages. Due to the imbalance of data points across classes, we propose the use of Stratified Group K-Folds to counteract this imbalance in class distribution. For our Deep Learning approach, we suggest aggressive geometric augmentation methods to significantly increase the dataset size due to rotational and translational invariance. We aim to accomplish this by rotating each image, and by implementing a sliding window which scans through the FOV images horizontally and vertically. It should be noted that we need to filter out and discard "empty" frames where not enough information is present in the image, as the original authors had done in their paper. The scale of these subimages must be selected to ensure that crucial spatial features are not compressed or distorted, and in a resolution which is optimal for whichever transfer learning architecture (e.g. 224×225 for ResNet, VGG, Inception, etc) is selected for this project.

Proposed Validation/Environment Strategy

The ability of our model to detect IDC-P in new, unseen patients will be tested by separating biopsy cores by patient during the training/testing selection process. I.e., biopsy cores from a selected number of patients to be used in evaluation to eliminate intra-core bias and to prevent the model from memorizing patient-specific features. This is an approach advised by the authors of the original paper.

Project Novelty

This project offers several novel elements:

1. Application of transferred-learned CNNs to multimodal nonlinear optical prostate imaging data.
2. Direct comparison between CNN-learned features and traditional texture-based methods.
   We intend to implement automatic feature selection using "end-to-end" Representational Learning and Regularization (e.g., L1 Lasso). The outcome will then be compared with the manual feature selection approach that the original authors used which involved using the Kruskal-Wallis test to prove discriminatory power with p-values and statistics.
3. Utilizing patient-level cross-validation in order to prevent data leakage and improve the evaluation reliability.
4. Interpretability analysis using Grad-CAM to identify the spatial regions. Ideally, our model focuses on generating a heatmap highlighting the most influential regions of an image that led to its prediction, improving the explainability of black box models.
5. Kernel architecture engineering and multimodal image co-registration.
   All three signals from SHG (Collagen), $1668\ cm^{-1}$ SRS (Protein / Amide I), and $1450\ cm^{-1}$ SRS (Lipids / Proteins) originate from the same laser focal volume; they provide perfectly aligned multi-channel inputs to a CNN. This allows us to experiment with various kernel architectures of depth 3 at the input layer.
6. Sliding window, Geometric Augmentation (rotational and translational invariance).
7. Stratified Group K-Fold for class distribution imbalance.

All of these contributions will help us compare the benefits and drawbacks to using deep learning on biomedical imaging classification as opposed to deterministic methods as presented in the original paper.

Milestone Schedule based on GitHub commits

Week	Milestone
March 2 - 8	Loading the dataset and working on preprocessing, verifying channel alignment
March 9 - 15	Implement the subimage extraction, data augmentation, and patient-level cross-validation
March 16 - 22	Implement the CNN architecture and establish the initial training baseline
March 23 - 29	Train the CNN models through pre-trained transfer learning models (RestNet-50, EfficientNet-B3)
March 30 - April 5	Implement multimodal fusion and evaluate CNN performance
April 6 - 8	Generate the Grad-CAM visualizations, analyze the results, and complete the final report

Resources:

https://www.nature.com/articles/s41598-025-23780-8